Examining the Risk of Substance Use and Lifestyle Factors on Cognitive Decline Among Older People in North-Eastern States of India

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Abstract

Objectives Increasing age and improper lifestyle negatively impact an individual's memory and cognition. Simultaneous use of substances and their detrimental effects such as cognitive impairment have been found among older adults. Maintaining healthy lifestyle habits in daily life may help prolong healthy cognition, and little work has been done on it in India. Thus, this paper inspects the key factors associated with cognitive impairment emphasizing smoked/smokeless tobacco and alcoholic beverages consumption among adults as a part of their lifestyle.

Method This study examines the association between substance use and lifestyle factors with cognitive decline among older adults (aged 45 years and above) in seven North-Eastern states of India using data from LASI Wave-1 (N = 7,790) . Binary logistic regression analysis was applied to understand the objectives.

Results The study estimate adjusted odds ratios (AOR) with 95% confidence intervals. Results indicate that advancing age, sleep problems, lack of reading habits, social disengagement, food insecurity, alcohol consumption, and urban residence were significantly associated with higher odds of cognitive impairment. Older adults who never consumed alcohol had 35% lower odds of impairment compared to alcohol users (AOR=0.65, $p<0.01$). Those with frequent sleep problems (≥5 nights/week) had more than three times higher odds of impairment (AOR=3.61, $p<0.01$).

Conclusion The study's findings aim to draw attention to detect cognitive impairment and suggest some recommended steps to reduce it among older individuals and emphasize the importance of addressing modifiable lifestyle factors to reduce cognitive decline in North-East India.. These also can be notified as public health initiatives to give quality care to elderly people.

Background

Population ageing is emerging as a critical public health concern across the North-Eastern region of India, where improvements in life expectancy have increased the proportion of older adults in recent decades. While longevity reflects progress in health and development, it also raises concerns about age-related morbidities, particularly cognitive decline. Cognitive decline, which includes deterioration in memory, attention, learning ability, and executive functioning, threatens independence, quality of life, and social participation among older individuals. In the North-Eastern states - comprising Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, and Tripura - the combined influence of geographic isolation, socio-economic disparities, limited health infrastructure, and changing lifestyle patterns may heighten vulnerability to cognitive impairment among the elderly. In LASI wave-1 percentage of consuming tobacco and alcohol is higher than other states only few states like Odisha has higher prevalence of tobacco consumption (64.0) than north-eastern states,also alcohol consumption is also high in north-eastern states where among men Manipur has highest prevalence (21.7) and for women Arunachal Pradesh has highest prevalence (5.2). So our study is based on only north-eastern states.

Substance use is one of the important yet under- examined determinants of cognitive health in later life. The North-Eastern region has distinct socio-cultural practices related to tobacco and alcohol consumption. Both smoked and smokeless tobacco are widely used, and traditional as well as commercially produced alcoholic beverages are socially embedded in several communities. Although moderate alcohol consumption has been debated in the literature for its potentially protective effects, heavy and prolonged use has consistently been associated with accelerated cognitive decline, memory impairment, and increased risk of dementia. Tobacco use, through vascular damage and neurotoxic effects, may also impair cognitive functioning over time. An older man who consumes heavy alcohol (≥36 grams/ day) has been associated with a faster decline in cognition in late middle age. Older adults, who often have coexisting chronic conditions such as hypertension and diabetes, may be particularly susceptible to the adverse cognitive consequences of substance use.

Beyond substance use, lifestyle factors play a significant role in shaping cognitive trajectories in later life. Engagement in physical activity, regular reading habits, and involvement in cultural or social programmes contribute to cognitive stimulation and social interaction, both of which are protective against cognitive deterioration. Conversely, physical inactivity, social isolation, and limited intellectual engagement may accelerate functional decline. In the North-Eastern context, disparities between rural and urban areas, variations in educational attainment, and differences in socio-economic status may further influence lifestyle behaviours and access to cognitively stimulating environments. Older adults living in rural and remote hilly terrains may face barriers to healthcare access, recreational facilities, and social engagement opportunities, thereby increasing their risk of cognitive impairment.

Socio-demographic characteristics such as age, caste, marital status, living arrangement, and economic status are also closely linked to cognitive outcomes. Advancing age remains the strongest predictor of cognitive decline, but its impact may be compounded by widowhood, living alone, poverty, and limited education. The North-Eastern states are marked by ethnic diversity and socio-cultural heterogeneity, which may shape health behaviours and health-seeking patterns differently compared to other regions of the country. Furthermore, chronic conditions such as hypertension and diabetes - both prevalent among older adults - are known risk factors for vascular cognitive impairment. When combined with substance use and unhealthy lifestyle practices, these conditions may intensify cognitive vulnerability.

Despite the growing burden of cognitive impairment, empirical evidence focusing specifically on the North-Eastern region remains limited. Most existing studies in India have examined cognitive decline at the national level, often overlooking regional heterogeneity. Given the unique socio-cultural patterns of substance consumption and lifestyle behaviours in the North-East, region-specific analysis is essential for developing targeted public health interventions.

Understanding how tobacco use,alcohol consumption, physical activity, and social engagement interact with socio-economic and health factors to influence cognitive outcomes can provide valuable insights for policy formulation.

Against this backdrop, the present study examines the risk of substance use and lifestyle factors on cognitive decline among older people in the North-Eastern states of India. The study hypothesizes that (1) consumption of smoked or smokeless tobacco and alcohol is positively associated with cognitive impairment among older adults, and (2) healthier lifestyle practices, including physical activity and social engagement, are associated with reduced odds of cognitive decline. By identifying modifiable risk factors within this region, the study aims to contribute to evidence-based strategies that promote healthy ageing and preserve cognitive well-being among the elderly population of North-East India.

Methods

Data

The present research applied data from the Longitudinal Aging Study of India (LASI) which was a nationally representative data conducted in 2017–2018 (wave 1) across all states & UTs of India. It is the world's largest and India's first longitudinal aging study that included over 72,000 older adults aged 45 years and above and their spouses irrespective of their age, however, the current study specifically focuses on the North-Eastern region.

For the purpose of this analysis, data were restricted to the eight North-Eastern states: Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, and Tripura. LASI employed a multistage stratified area probability cluster sampling design. In rural areas, a three-stage sampling procedure was adopted, involving the selection of Primary Sampling Units (PSUs) such as sub-districts (tehsils/talukas), followed by villages, and finally households. In urban areas, a four-stage sampling design was used, which included the additional selection of Census Enumeration Blocks (CEBs) before selecting households. This design ensures representativeness at both state and regional levels.

While LASI Wave 1 interviewed individuals aged 45 years and above, the present study focuses exclusively on older adults aged 60 years and above residing in the North-Eastern states. After restricting the dataset to this age group and excluding cases with missing information on key variables, the final analytical sample consisted of 7,790 older individuals because our study is only consider north-eastern states. This region-specific subsample provides a robust basis for examining the association between substance use, lifestyle factors, and cognitive impairment among older adults in North-East India.

Variable description

Outcome variables

During the time of collecting the information regarding cog nitive impairment in LASI data, the interviewer read out a list of 10 words, and individual respondent was asked to repeat the words using their memory. A scale of 0 to 10 was prepared for this measurement. A higher score means lower cognitive impairment and vice-versa. However, the report did not include the words used for measuring cognitive impairment. Respondents who can recall five or more words recoded as 0 "low" representing lower cognitive impairment and a score of four or less recoded as 1 "high" representing higher cognitive impairment.

Explanatory variable

There are two main explanatory variables to fulfill the study objectives. 1. Smoke/smokeless tobacco was recoded as 0 "no" and 1 "yes". 2. Consumed any alcoholic beverages was recoded as 0 "no" and 1 "yes". The questions appraised 'Ever smoked or used smokeless tobacco' and 'Ever consumed any alcoholic beverage.'

Age was recoded 45-59 as "Adult" ,60–79 as "Older adult" and 80+ years as "Oldest adult", Sex was recoded as male and female. Ever attend school was recoded as "yes" and "no." The residence was recoded as urban and rural. Caste was recoded as SC/ST and non-SC/ST "included OBC and none of them". Living arrangement was recoded as living with someone "included spouse, children, and others" and living alone. Marital status was recoded as currently married, live in relationship, and not in union "included widowed, divorced, separated and deserted". Attend cultural performance was recoded as sometimes in a month, rarely, and never. Eat food of your own choice was recoded as binary form 0 "yes" and 1 "no." Reading newspapers or magazines or books was recoded as daily, sometimes, and never. Doing yoga or pranayama or medication or asanas were recoded as every day, several times in a month, and hardly ever or never. Trouble sleeping problem was recoded as never, 1–2 nights/week, 3–4 nights/ week, 5 or more nights/week. The wealth index was recoded as poorest , poorer , middle, richer, richest .Table 1 highlights all the codes of the categorical variables.

Statistical analysis

Descriptive statistics as well as percentage distribution were calculated for cognitive impairment over explanatory variables. To measure the significance level between the outcome and explanatory variable, the Chi-square test of association was applied. To find more association between cognitive impairment over explanatory variables binary logistic regression model (Osborne 2008) was run through STATA 17. The outcome variable was cognitive impairment which was coded as "low (0) and "high" (1). In this study, consumption of tobacco (smoked/smokeless) and drinking alcoholic beverages (wine, tadi, country liquor, beer, and arrack) were the main explanatory variable. The regression for logistic distribution is as follows: $\ln\left[\frac{\Pi}{1-\Pi}\right] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_n x_n$ where $\beta_0, \ldots$ βn, are the intercepts regression coefficients measuring the relative effect of particular explanatory vari ables on mental cognition and ∈ represents the residual or unexplained variation in the model. With the help of an adjusted odd ratio (AOR) with a 95% confidence interval (CI), the results were presented.

Results

Table 2 presents the socio-demographic and lifestyle profile of adults (aged 45 years and above) from the selected North-Eastern states of India. Slightly more than half of the respondents reported ever using smoked or smokeless tobacco (50.3%), while 24.7% reported alcohol consumption. A majority of the sample belonged to the adult age group 45-59 (55.5% adults), followed by older adults (38.4%) and oldest-old (6.1%). Females constituted 52.4% of the study population. Nearly 40% of respondents had never attended school, and about three-fourths (74.2%) resided in rural areas. Most respondents were currently in union (73.5%), while 26.5% were not in union. A substantial proportion belonged to the SC/ST community (71.8%) and rest (28.2%) are from other caste such as OBC and others, and the majority (96.0%) were living with someone.

Regarding socioeconomic status, the distribution across wealth quintiles was relatively even, although a slightly higher share was observed among the richest (22.9%) follow with richer (20.78%) middle with (19.73%) poorer (18.75%) and rest are poorest (17.87%). In terms of lifestyle behaviors, 63.6% of older adults never attended any cultural programme, and 68.7% reported almost never engaging in reading activities follow with 18.87% reported daily reader and rest 12.41% reads sometimes. Physical inactivity was highly prevalent, with 86.9% reporting that they never performed physical exercise. Additionally, 53.1% reported that they did not eat enough food of their own choice. Sleep problems were also common: while 57.3% reported no trouble falling asleep, approximately 42.7% experienced sleep disturbances at least once a week.

Consistent with findings reported in the Results section of Saha et al.(2025) , the distribution of lifestyle factors indicates a high prevalence of behavioral and social risk factors that are associated with cognitive impairment among older adults. The descriptive profile suggests that substance use, limited social participation, lack of reading habits, physical inactivity, and sleep disturbances are common among the elderly population in the North-East. Given that these factors have been shown to significantly increase the likelihood of cognitive decline in adjusted models, the observed patterns highlight a substantial vulnerability among older adults in this region. Overall, the results underscore the importance of addressing modifiable lifestyle and behavioral determinants to mitigate the risk of cognitive impairment among ageing populations in North-Eastern India.

Table 3 presents the binary logistic regression estimates of factors associated with cognitive impairment among older adults aged 45 years and above in the North-Eastern states of India (N = 7,790). Odds ratios (OR) with 95% confidence intervals (CI) are reported.
Advancing age demonstrated a strong and statistically significant association with cognitive impairment. Compared to adults aged 45–59 years, older adults had significantly higher odds of impairment [OR 1.683; CI 1.289–2.197; $p < 0.001$], while the oldest-old exhibited nearly three times higher odds [OR 2.999; CI 2.012–4.472; $p < 0.001$]. This finding confirms a clear age-gradient effect in cognitive vulnerability.

Caste did not show a statistically significant association in the adjusted model. Non-SC/ST respondents had higher odds relative to SC/ST, but the association was not significant [OR 1.161; CI 0.892–1.511; $p = 0.268$].
Regarding physical activity, respondents who never engaged in physical activity had significantly lower odds compared to those performing daily activity [OR 0.684; CI 0.478–0.979; $p < 0.05$], although the direction suggests complexity in interpretation. Participation several times per month was not statistically significant [OR 0.640; CI 0.222–1.848; $p = 0.410$].
Education showed a significant association. Individuals with no schooling had lower odds compared to those who attended school [OR 0.695; CI 0.521–0.928; $p < 0.05$]. This indicates that educational status remains an important determinant in the adjusted framework.
Reading habits were not statistically significant in the multivariable model. Compared to regular readers, those who read sometimes [OR 1.003; CI 0.619–1.626; $p = 0.989$] and those who almost never read [OR 1.326; CI 0.912–1.929; $p = 0.139$] did not show meaningful differences.
Attendance at cultural or social programmes showed partial significance. Compared to those attending often, respondents who attended rarely had significantly lower odds of cognitive impairment [OR 0.398; CI 0.211–0.747; $p < 0.01$], whereas those who never attended did not show a statistically significant association [OR 0.785; CI 0.433–1.424; $p = 0.426$].
Living arrangement was not associated with cognitive impairment. Individuals living with someone had similar odds compared to those living alone [OR 0.974; CI 0.564–1.681; $p = 0.923$].
Marital status demonstrated a strong association. Compared to married individuals, those currently not in union had significantly higher odds of cognitive impairment [OR 1.899; CI 1.427–2.526; $p < 0.001$]. Although individuals in live-in relationships had elevated odds [OR 7.158; CI 0.511–100.161], the wide confidence interval suggests instability likely due to small sample size.
Food insecurity emerged as a significant predictor. Respondents who reported no food insecurity had higher odds relative to those experiencing food insecurity [OR 1.379; CI 1.074–1.769; $p < 0.05$], indicating that nutritional and socio-economic factors may influence cognitive outcomes.
Wealth index did not retain statistical significance in the adjusted model. Compared to the poorest group, poorer [OR 1.137; CI 0.780–1.658], middle [OR 0.948; CI 0.643–1.396], richer [OR 0.837; CI 0.562–1.248], and richest [OR 0.921; CI 0.623–1.362] categories did not show meaningful differences.
Sex was not significantly associated with cognitive impairment. Females had slightly lower odds compared to males [OR 0.904; CI 0.670–1.218; $p = 0.506$].
Sleep disturbance demonstrated one of the strongest associations in the model. Compared to respondents who never experienced trouble sleeping, those reporting sleep problems rarely (1–2 nights/week) had significantly higher odds [OR 1.700; CI 1.288–2.243; $p < 0.001$]. The odds increased substantially among those reporting occasional sleep disturbance (3–4 nights/week) [OR 3.107; CI 2.211–4.367; $p < 0.001$], and frequent sleep disturbance (≥5 nights/week) [OR 3.602; CI 2.105–6.164; $p < 0.001$]. This graded increase suggests a dose–response relationship between sleep problems and cognitive impairment.
Substance use variables showed differential effects. Tobacco use was not significantly associated after adjustment. Individuals who never smoked had similar odds compared to smokers [OR 0.879; CI 0.679–1.138; $p = 0.329$]. In contrast, alcohol consumption demonstrated a statistically significant association. Respondents who never consumed alcohol had 37.5% lower odds of cognitive impairment compared to alcohol users [OR 0.625; CI 0.462–0.846; $p < 0.01$].
Place of residence did not reach statistical significance, although urban residents had somewhat higher odds compared to rural residents [OR 1.243; CI 0.935–1.652; $p = 0.135$].
Overall, the findings indicate that advancing age, marital disruption, food insecurity, sleep disturbance, and alcohol consumption are significant determinants of cognitive impairment among older adults in the North-Eastern states of India. Among these factors, age and sleep disturbance emerged as the strongest predictors in the adjusted model.

From table 4 which is basically showing the unadjusted ORs represent the independent association between each explanatory variable and cognitive impairment without controlling for other covariates. In contrast, the adjusted ORs account for the simultaneous influence of multiple socio-demographic, lifestyle, and substance use variables. As a result, the adjusted model provides a more reliable estimate of the net effect of each predictor.

Several associations changed after adjustment. For instance, some variables that appeared important in crude analysis lost statistical significance in the adjusted model, such as residence, caste, and certain lifestyle behaviors. This suggests that their initial associations were confounded by other correlated factors like age, marital status, or sleep problems. Conversely, key predictors such as advancing age, sleep disturbance, marital disruption, and showed progressively increasing odds across categories in the adjusted model, while alcohol non-consumption retained a protective effect.

Overall, the adjusted ORs refine the interpretation by distinguishing true independent predictors from associations explained by other variables. Therefore, Table 3 provides controlled estimates, whereas Table 4's crude ORs reflect simple bivariate relationships.

When ORs change between crude and adjusted models, it means that there was confounding ,variables were correlated, some effects were partially explained by other predictors in example residence may look significant in crude analysis but once education, wealth, and lifestyle are included, it becomes non-significant.

Variables like age , sleep disturbance , alcohol consumption , marital status remain significant after adjustment because their effect is strong and they independently predict impairment.

The adjusted odds ratios provide a more valid estimate of the determinants of cognitive impairment because they account for interrelationships among age, socio-demographic factors, lifestyle behaviors, and substance use. Differences between crude and adjusted ORs indicate the presence of confounding, while stability of ORs across models suggests robust independent associations.

From the graph 1 we can say that the predicted probability of cognitive impairment increases steadily with age for both tobacco users and non-users, confirming a strong age effect. In each age group, tobacco users have a slightly higher predicted probability of impairment compared to non-users, indicating a positive association between tobacco use and cognitive decline. However, the 95% confidence intervals (CI) around the estimates overlap across most age categories, suggesting that the difference between users and non-users is not very large. The confidence intervals become wider in the oldest age group, indicating less precision in the estimates, possibly due to smaller sample size or greater variability in that group also we can infer that those adults who are in higher age group and consume tobacco has higher possibilities of having cognitive impairment.

The graph shows that the predicted probability of cognitive impairment increases clearly with age among both alcohol users and non-users, confirming a strong age effect. Across all age groups, alcohol users consistently have a higher predicted probability of impairment compared to those who never consumed alcohol. Unlike the tobacco graph, the gap between the two groups is more noticeable, especially among the oldest adults. The 95% confidence intervals (CI) show less overlap between alcohol users and non-users, particularly in older age groups, suggesting a stronger and more statistically meaningful association. The wider CIs in the oldest group indicate lower precision but still support a significant age-related increase in risk also it can be said that for consuming alcohol it has a significant effect on having cognitive decline among oldest age group.

Discussion

The present study examined the association between substance use, lifestyle factors, and cognitive impairment among adults aged 45 years and above in the North-Eastern states of India. Consistent with our first hypothesis, alcohol consumption was significantly associated with higher odds of cognitive impairment, even after adjusting for socio-demographic and behavioral covariates. This finding aligns with earlier evidence suggesting that substance use contributes to adverse neurological outcomes and accelerates cognitive decline in later life (Moore et al. 2006; Xu et al. 2009). Although tobacco use did not retain statistical significance in the fully adjusted model, the direction of association was consistent with prior studies indicating that long-term exposure to tobacco may increase morbidity and impair cognitive functioning (Prakash et al. 2004).

Advancing age emerged as the strongest predictor of cognitive impairment, confirming that cognitive vulnerability increases progressively with aging. This observation is consistent with previous literature demonstrating that declining cognitive function is closely linked with biological aging and heightened risk of chronic disease and mortality (Yaneva-Sirakova and Traykov 2022). Marital disruption also showed a significant association, as individuals currently not in union had higher odds of impairment. This supports earlier research suggesting that widowhood and social instability may heighten emotional stress and increase engagement in unhealthy behaviors (Håkansson et al. 2009).

Sleep disturbance demonstrated one of the most robust associations, with a clear dose–response pattern. Older adults reporting frequent trouble falling asleep had substantially elevated odds of cognitive impairment. These findings corroborate earlier studies that documented strong positive correlations between sleep disorders and reduced cognitive performance (Cricco et al. 2001; Yaffe et al. 2014). Poor sleep quality may impair memory consolidation and executive functioning, thereby accelerating cognitive decline.

Socio-economic factors also played an important role. Food insecurity was significantly associated with cognitive impairment, indicating that nutritional vulnerability may adversely affect mental functioning. Although wealth index and caste were not statistically significant in the adjusted model, earlier research has highlighted the influence of socio-economic disadvantage on cognitive health (Muhammad et al. 2021, 2022).

Lifestyle engagement showed mixed effects. Attendance at cultural programmes exhibited partial protective association, reflecting the broader understanding that social ties and participation promote psychological well-being (Child and Albert 2018; Strawbridge et al. 2001). Physical inactivity and lack of reading habits, although not uniformly significant, followed patterns observed in previous studies where regular exercise and cognitive stimulation improved mental functioning (Clarkson-Smith and Hartley 1989; Gheysen et al. 2018; Sörman et al. 2018).

This study has several limitations. First, the cross-sectional design limits causal inference between substance use and cognitive impairment. Second, LASI data were collected during 2017–2018, and patterns may have changed over time. Third, cognitive impairment was measured primarily through recall-based assessments and did not comprehensively capture executive or orientation domains. Additionally, substance use patterns such as duration and intensity were not examined. Despite these limitations, the study provides region-specific evidence for the North-Eastern states, highlighting modifiable behavioral and social determinants of cognitive health.

Conclusion

This study highlights the growing burden of cognitive impairment among adults aged 45 years and above in the North-Eastern states of India, The findings confirm that advancing age, alcohol consumption, sleep disturbances, marital disruption, and food insecurity are significant determinants of cognitive decline in this region. In particular, older adults who consumed alcohol and those experiencing frequent sleep problems demonstrated substantially higher odds of impairment, underscoring the role of modifiable behavioral factors.

The results further suggest that limited social engagement and poor socio-economic conditions may exacerbate cognitive vulnerability among ageing populations. While aging is an inevitable biological process, lifestyle behaviors such as alcohol use, poor sleep hygiene, and reduced social participation represent preventable risk factors.

Therefore, early screening of cognitive function, community-level awareness programs, and targeted public health strategies focusing on substance use reduction and healthy ageing practices are essential. Addressing these modifiable determinants can contribute to preserving cognitive well-being and improving quality of life among older adults in North-East India.

Table 1: Description of Variables and Associated Coding

Variables	**Associated coading**
Cognitive impairment	Low = 0; High= 1
Smoked/smokeless tobacoo	Yes = 0 ; No = 1
Alcoholic beverages consumption	Yes = 0 ; No = 1
Age	45-59 = 0 ; 60-79 = 1 ; 80+ = 2
Sex	Male = 1; Female = 2
Attend school	Yes = 0 ; No = 1
Residence	Rural = 1 ; Urban = 2
Marital status	Currently in union = 1 ; Live in relationship = 2 ; Currently not in union = 3
Caste	Non SC/ST = 1 ; SC/ST = 2
Living arrangements	Living with someone = 1 ; Living alone = 2
Wealth index	Poorest = 1 ; Poorer = 2 ; Middle = 3 ; Richer = 4 ; Richest = 5
Attend cultural performance	Sometimes in a month = 1 ; Rarely = 2 ; Never = 3
Eat enough food of your own choice	Yes = 0 ; No = 1
Resding habits	Daily = 1 ; Sometimes = 2 ; Never = 3
Doing physical exercise	Everyday = 1 ; Several times in month = 2 ; Never = 3
Trouble falling asleep	Never =1 ; 1-2 nights/week = 2 ; 3-4 nights/week = 3; 5 or more days/week = 4

Table 2: Socio-demographic, Substance Use and Lifestyle Characteristics of Older Adults in North-East India (N = 7,790) (Chi-square test)

Variable	p <0.05	Sample	Percentage
Smoked/smokeless tobacco	*		
No		3872	49.70
Yes		3918	50.30
Alcoholic beverages consumption	*		
No		5864	75.28
Yes		1926	24.72
Age (Years)	*		
Adult		4324	55.51
Older Adult		2994	38.43
Oldest Adult		472	6.06
Sex of Respondent			
Male		3712	47.65
Female		4078	52.35
Ever attended school			
Yes		4648	59.67
No		3142	40.33
Residence			
Rural		5779	74.18
Urban		2011	25.82
Marital Status	*		
Currently in union		5722	73.45
Live in relationship		4	0.05
Currently not in union		2064	26.50
Caste	*		
SC/ST		5591	71.77
Non-SC/ST		2199	28.23
Living arrangements			
Living alone		310	3.98
Living with someone		7480	96.02
Wealth index			
Poorest		1392	17.87
Poorer		1461	18.75
Middle		1537	19.73
Richer		1619	20.78
Richest		1781	22.86
Attend cultural programme	*		
Several times in a month		230	2.95
Rarely/since a year		2608	33.48
Never		4952	63.57
Eat enough food of your choice	*		
Yes		3653	46.89
No		4137	53.11
Reading habit			
Almost Daily		1470	18.87
Sometimes		967	12.41

Almost Never		5353	68.72
Physical exercise			
Almost everyday		889	11.41
Several times in month		130	1.67
Never		6771	86.92
Trouble falling asleep	*		
Never		4465	57.32
Rarely (1–2 nights/week)		2437	31.28
Occasionally (3–4 nights/week)		712	9.14
Frequently (5+ nights/week)		176	2.26

Note: Percentages are calculated from total sample size (N = 7,790). All percentages within each variable sum to 100%. SC/ST Scheduled Caste/Schedule Tribe; *if $p < 0.05$

Table 3: Binary Logistic Regression Analysis of Factors Associated with Cognitive Impairment Among Older Adults in North-East India (N = 7,790)

Variable	Category	OR	95% CI	p-value
Age (ref: Young adult)				
	Older adult	1.683***	1.289 – 2.197	<0.001
	Oldest adult	2.999***	2.012 – 4.472	<0.001
Caste (ref: SC/ST)				
	Non SC-ST	1.161	0.892 – 1.511	0.268
Physical activity (ref: Daily)				
	Several times/month	0.640	0.222 – 1.848	0.410
	Never	0.684**	0.478 – 0.979	0.038
Education (ref: Attended school)				
	No schooling	0.695**	0.521 – 0.928	0.014
Reading habit (ref: Regular)				
	Sometimes	1.003	0.619 – 1.626	0.989
	Almost never	1.326	0.912 – 1.929	0.139
Attend programme (ref: Often)				
	Rarely	0.398***	0.211 – 0.747	0.004
	Never	0.785	0.433 – 1.424	0.426
Living arrangement (ref: Alone)				
	Living with someone	0.974	0.564 – 1.681	0.923
Marital status (ref: Married)				
	Live-in relationship	7.158	0.511 – 100.161	0.144
	Currently not in union	1.899***	1.427 – 2.526	<0.001
Food insecurity (ref: Yes)				
	No	1.379**	1.074 – 1.769	0.012
Wealth index (ref: Poorest)				
	Poorer	1.137	0.780 – 1.658	0.504
	Middle	0.948	0.643 – 1.396	0.786
	Richer	0.837	0.562 – 1.248	0.383
	Richest	0.921	0.623 – 1.362	0.681
Sex (ref: Male)				
	Female	0.904	0.670 – 1.218	0.506
Sleep trouble (ref: Never)				
	Rarely (1–2 nights/week)	1.700***	1.288 – 2.243	<0.001

Occasionally (3–4 nights/week)	3.107***	2.211 – 4.367	< 0.001
Frequently (≥5 nights/week)	3.602***	2.105 – 6.164	< 0.001
Ever smoked tobacco (ref: Yes)			
No	0.879	0.679 – 1.138	0.329
Ever consumed alcohol (ref: Yes)			
No, never	0.625***	0.462 – 0.846	0.002
Place of residence (ref: Rural)			
Urban	1.243	0.935 – 1.652	0.135
Constant	0.038***	0.014 – 0.102	< 0.001

Notes: Odds Ratios (OR) reported. Reference categories shown in parentheses. *** p <0.01, ** p <0.05, * p < 0.10.
Model statistics: LR $\chi^2 = 200.31$; Pseudo $R^2 = 0.080$; AIC = 2355.34; BIC = 2536.32.

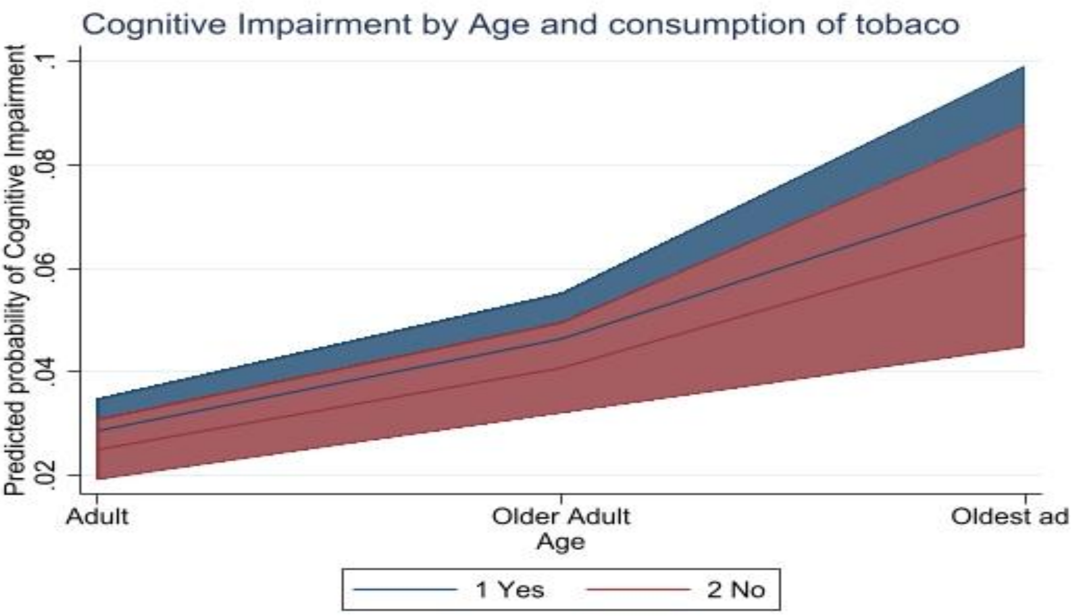
Table 4: Unadjusted Odds Ratios from Logistic Regression Model

Variable	**Category**	**OR**	**SE**	**z**	**p-value**	**95% CI**
Tobacco use	No (Ref.)	—	—	—	—	—
	Yes	0.729	0.088	−2.62	0.009	(0.576, 0.923)
Alcohol use	No (Ref.)	—	—	—	—	—
	Yes	0.735	0.095	−2.38	0.017	(0.571, 0.947)
Trouble falling asleep	Never (Ref.)	—	—	—	—	—
	Rarely (1–2 nights/wk)	1.818	0.251	4.33	0.000	(1.387, 2.383)
	Occasionally (3–4 nights/wk)	3.511	0.588	7.49	0.000	(2.528, 4.877)
	Frequently (5+ nights/wk)	4.791	1.252	5.99	0.000	(2.871, 7.997)
Age group	Adult (Ref.)	—	—	—	—	—
	Older Adult	1.995	0.260	5.30	0.000	(1.546, 2.577)
	Oldest Adult	4.194	0.776	7.75	0.000	(2.918, 6.027)
Sex	Male (Ref.)	—	—	—	—	—
	Female	1.001	0.119	0.01	0.991	(0.793, 1.264)
Ever attended school	Yes (Ref.)	—	—	—	—	—
	No	1.001	0.119	0.01	0.991	(0.793, 1.264)
Residence	Rural (Ref.)	—	—	—	—	—
	Urban	1.176	0.154	1.24	0.217	(0.909, 1.521)
Marital status	In union (Ref.)	—	—	—	—	—
	Live-in relationship	11.227	12.994	2.09	0.037	(1.162, 108.499)
	Currently not in union	2.227	0.269	6.63	0.000	(1.758, 2.821)
Caste	SC/ST (Ref.)	—	—	—	—	—
	Non-SC/ST	1.264	0.160	1.85	0.065	(0.986, 1.620)
Living arrangements	Alone (Ref.)	—	—	—	—	—
	With someone	0.663	0.170	−1.60	0.109	(0.401, 1.097)

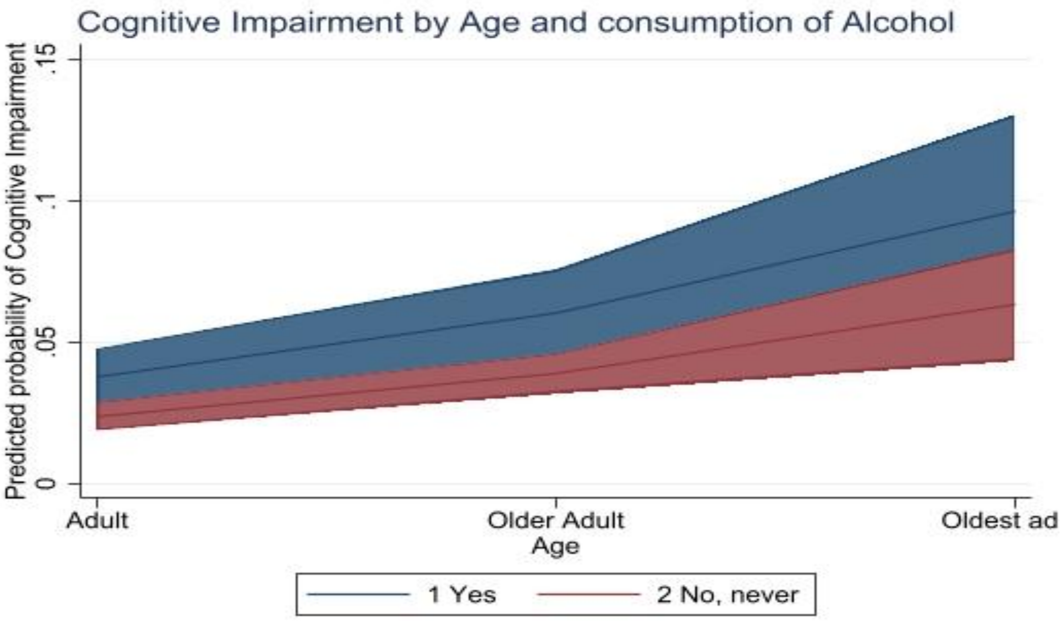
Wealth index	Poorest (Ref.)	—	—	—	—	—
	Poorer	1.132	0.212	0.66	0.508	(0.784, 1.633)
	Middle	0.970	0.185	−0.16	0.875	(0.667, 1.411)
	Richer	0.823	0.161	−1.00	0.320	(0.560, 1.208)
	Richest	0.877	0.165	−0.70	0.486	(0.605, 1.269)
Cultural programme	Several times/month (Ref.)	—	—	—	—	—
	Rarely/since a year	0.393	0.123	−2.97	0.003	(0.212, 0.727)
	Never	0.779	0.780	−0.85	0.397	(0.397, 1.438)
Adequate food of choice	Yes (Ref.)	—	—	—	—	—
	No	1.371	0.167	2.60	0.009	(1.080, 1.740)
Reading habit	Almost daily (Ref.)	—	—	—	—	—
	Sometimes	1.014	0.243	0.06	0.954	(0.634, 1.621)
	Almost never	1.351	0.225	1.81	0.071	(0.975, 1.872)
Physical exercise	Almost everyday (Ref.)	—	—	—	—	—
	Several times/month	0.691	0.369	−0.69	0.490	(0.243, 1.969)
	Never	0.839	0.148	−1.00	0.319	(0.594, 1.185)

Note: OR = Odds Ratio; SE = Standard Error; CI = Confidence Interval. Ref = Reference category.

Graph 1



Graph 2



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